Shahriar Alom Masud

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LinkedIn | GitHub | Portfolio

Professional Summary

Embedded Software and IoT Engineer with 2+ years of hands-on experience developing C/C++ firmware for microcontrollers. Skilled in FreeRTOS, dual-partition OTA updates, wireless sensor communication, and running lightweight machine learning models on edge devices.

Skills

Languages: C, C++, Python, JavaScript, TypeScript, Dart, SQL, Bash

Embedded: ESP32, ESP8266, Raspberry Pi, Arduino, FreeRTOS, Dual-Partition OTA, NVS, EEPROM, Hardware Timers, ISR

Protocols: MQTT, HTTP/HTTPS, LoRa, BLE, UART, SPI, I2C, 1-Wire, WebSockets

AI / ML: TinyML, ONNX Runtime, OpenCV, YOLOv8, PyTorch, Scikit-learn, MediaPipe

Cloud & Tools: Azure, FastAPI, Supabase, Firebase, MongoDB, InfluxDB, Docker, React, Flutter, Git, GitHub Actions, PlatformIO, Linux

Experience

Open Source Developer — SmartProv *Apr 2026 – Present*

github.com/Masud744/SmartProv — *Arduino Library Manager, ESP32, ESP8266, C++, NVS/EEPROM*

- Built and published **SmartProv** on the Arduino Library Manager — an open-source Wi-Fi provisioning library for ESP32 and ESP8266 that lets users configure device credentials through a captive portal instead of reflashing firmware.
- Stores Wi-Fi credentials in NVS and EEPROM with multi-network fallback and factory-reset support, so devices reconnect automatically after power loss without manual reflashing.
- Cuts device network setup time from several minutes to under 30 seconds per node, removing the need to open Arduino IDE for every Wi-Fi change.

Projects

RespiGuard.ai — Cyber-Physical Edge AI Respiratory Monitoring *May 2026 – Sep 2026*

github.com/Masud744/RespiGuard — Live: respiguard-fvh9.onrender.com

ESP32, FreeRTOS, C/C++, PMS5003 (UART), MQ-135 (ADC), DHT22, SSD1306 OLED, FastAPI, TreeSHAP, React

- Built a multi-sensor ESP32 edge node reading laser particulate matter ($PM_{1.0}$/2.5/10 via UART2), calibrated toxic gas (MQ-135 via 12-bit ADC1), and climate data using FreeRTOS tasks and SSD1306 I2C display.
- Implemented dual-pipeline edge/clinical inference with TreeSHAP explainability to evaluate asthma risk zones (Green, Yellow, Red based on PEFR), streaming HMAC-SHA256 authenticated telemetry over HTTPS to a FastAPI and React dashboard.

Smart Solar-Grid HEMS — Intelligent Load Scheduling & Power Routing *Jan 2026 – Sep 2026*

github.com/Masud744/solar-aware-hems — Live: solar-aware-hems.vercel.app

ESP32, C++, FreeRTOS, ZMPT101B, ACS712, 8-Relay Matrix, NVS Flash, Python, FastAPI, Scikit-learn, React

- Engineered dual-core ESP32 firmware running a 1.5–2.0 kHz True-RMS AC sampling loop on Core 1 (ZMPT101B voltage, ACS712 current with NVS calibration) and Wi-Fi cloud telemetry on Core 0.
- Designed an 8-relay dual-bank matrix (Grid vs Solar) with 300 ms break-before-make dead-time protection and anti-chatter hysteresis, scheduling loads using a $k \times \sigma$ risk-aware solar forecasting engine.

ESP32 GitHub OTA — Fail-Safe Firmware Update & Recovery System *Jun 2026 – Aug 2026*

github.com/Masud744/esp32-smartprov-github-ota

ESP32, C/C++, Dual-Partition Flash (ota_0/ota_1), NVS, HTTPS, GitHub REST, PlatformIO

- Developed a serverless OTA client on ESP32 that checks GitHub-hosted `version.json` over HTTPS and streams compiled binaries into inactive flash partitions with automatic download retry limits.
- Built crash-recovery logic using NVS boot counters; automatically triggers `Update.rollBack()` if a new build crashes 3 times or drops Wi-Fi, blacklisting faulty releases to prevent infinite boot-loops.

Education

University of Frontier Technology, Bangladesh *Aug 2023 – Jun 2027 (Expected)*

Bachelor of Science (Engineering) in Internet of Things & Robotics Engineering *Gazipur, Bangladesh*

Leadership

Head of Membership Development — UFTB Robotics Club *Oct 2025 – Present*

Organized hands-on technical workshops on microcontroller interfacing, C/C++ firmware debugging, and sensor-to-cloud telemetry for 60+ junior engineering students.

Robo Soccer Segment Leader — RoboFusion 1.0 *Aug 2026*

Led and coordinated the Robo Soccer segment across 20 participating inter-university teams, overseeing arena technical setups, wireless frequency allocations, and match execution.